

Results: what resolved and what stayed
bounded

Results: what resolved and what stayed bounded I

Numbers below are token-injected by `src/ntqr_allotment/manuscript_variables.py` from the artifact named in each section: `sweep_aggregated.csv` for strategy and size summaries, `independence_sweep.csv` for tolerance, `postdoc_panel_results.json` and `postdoc_panel_alignment.json` for the live Gemma reviewer-panel companion, `power_analysis.csv` / `sweep_results.json` for design budgets, and `alarm_timings.csv` for alarm timing. None are hand-transcribed. Errors are mean EIE recovery error against the supervised oracle, aggregated over 96 seeds in the active sweep profile.

Results: what resolved and what stayed bounded II

This section adjudicates the five hypotheses from the Introduction. H1, H3, and H5 each resolve in one subsection. H2 and H4 are addressed in two stages: first against the baseline grid (where, by construction, both are design-limited because the baseline judges err independently), then resolved together by the composition-coupled confound. Remaining subsections report supporting diagnostics. The Discussion returns the per-hypothesis verdicts.

Synthetic deterministic results: controlled spine for H1-H4 I

The synthetic results are generated from seeded populations and corpora with a known oracle. They support strategy, regime, tolerance, power-budget, alarm, and diagnostic claims for the active deterministic profile only. Throughout, a *regime* is one (expert-stringency $\times$ bias-spread $\times$ panel-size) cell of the sweep grid (sixteen cells in the active profile); contrasts are evaluated cell by cell and averaged only where the text says so.

Synthetic deterministic results: controlled spine for H1-H4

Formation strategy sets the recovery floor

H1 (formation strategy is the dominant lever). The strategies do not fan out into a graded four-way ranking; instead one rule stands far apart and the other three collapse together ([@fig:ranking]). Competence-first selection recovers at mean EIE error 0.037 — roughly a quarter of the error of any other rule — while representative sortition, random selection, and single-bloc ideological selection are **statistically indistinguishable from one another**, clustered at 0.147–0.148:

Strategy	Mean EIE error 95% CI	
expertise threshold (far best)	0.037	± 0.003
representative sortition	0.147	± 0.014
random selection	0.148	± 0.014
single-bloc ideological selection	0.148	± 0.015

Formation strategy sets the no-answer-key error ceiling



Real-Ollama postdoctoral panel results: live H5 companion I

The live-Ollama results are separate empirical companion artifacts. They use one local gemma3:4b model prompted as synthetic postdoctoral reviewers over fictitious applications with synthetic age metadata. The result is not a model-family comparison, not a human-review validation, and not evidence that age belongs in real admissions or hiring review.

Real-Ollama postdoctoral panel results: live H5 companion

II Gemma ranking asks the same sampling question under prompt labels

The live artifact uses 12 seeds, 48 reviewers, 72 applications per seed, panel sizes 3, 6, and live Ollama provenance (gemma3:4b digest a2af6cc3eb7f, config hash 5161ffe474b3, vote-cache entries 23688). The best live postdoc EIE point estimate is same-bias selection at 0.216; the worst is expertise threshold at 0.347. For the three-seat panels, representative sortition has EIE 0.225, same-bias selection has 0.228, expertise-threshold selection has 0.347, and random selection has 0.262.

